

TAYLOR MCDONALD

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DATA SCIENCE INTERN | ML ENGINEERING

Building ML-Powered Systems at the Intersection of Data, Business, and Real-World Impact

Data Science student at NJIT with proven ability to build and deploy end-to-end ML systems on real-world data. Designed full-stack pipelines spanning EDA, feature engineering, model training, REST API deployment, and interactive dashboards. Founded All The Paws (15+ recurring clients) and built PawsIQ on top of that operational data. Strong quantitative foundation in statistical modeling. Seeking to contribute technically from day one in a fast-paced, impact-driven environment.

TECHNICAL SKILLS

Python | SQL | scikit-learn | XGBoost | pandas | numpy | FastAPI | React | Git | Jupyter Notebook | Docker | Vercel

CORE COMPETENCIES

Machine Learning | Statistical Modeling | Regression Analysis | EDA | Feature Engineering | Data Pipelines | Hyperparameter Tuning
Model Evaluation | Model Deployment | REST APIs | Data Storytelling | Cross-functional Communication | Entrepreneurship

PROJECTS

PawsIQ – Pet Services Intelligence Platform (In Progress) | April 2026 – Present

Python | scikit-learn | pandas | FastAPI | React | Vercel

- Performed EDA on 9,935 NJ bookings to identify demand patterns by time, location, and walker; developed GradientBoosting forecasting model achieving 5% RMSE improvement with stable cross-validation MAE of 0.774 ± 0.033 across 5 time-series folds.
- Engineered dynamic pricing pipeline using Ridge Regression ($R^2=0.727$) to predict surge multipliers per booking, producing 16.7% simulated revenue lift over flat-rate baseline across 1,856 held-out test bookings.
- Designed FastAPI backend serving both ML models as live REST endpoints; built React operator dashboard visualizing demand heatmaps, surge pricing schedules, and walker performance across 8 walkers.

House Prices Prediction | March 2026

Python | pandas | numpy | scikit-learn | matplotlib | seaborn

- Performed EDA across 79 features of a residential housing dataset; engineered 5 predictive variables using Random Forest importance, reducing noise and improving model signal.
- Trained and compared Linear Regression, KNN, and Random Forest on 1,460 samples, achieving $R^2=0.87$ and $RMSE=0.14$; tuned hyperparameters ($n_estimators$, max_depth) via cross-validation for 22% RMSE improvement over baseline.
- Produced a fully reproducible data pipeline from raw Kaggle CSV to final predictions with documented methodology and assumptions.

WORK EXPERIENCE

SETI Institute – Undergraduate Research Intern | Remote | Jun 2026 – Aug 2026

- Contributing to an NSF-funded research project developing machine learning models to forecast summertime Arctic sea ice extent and regional coverage from weeks to a full season ahead
- Analyzing large observational datasets including satellite, reanalysis, and global climate records to capture non-linear interactions across local and remote climate drivers
- Applying Python and MATLAB-based ML pipelines in an interdisciplinary research environment in collaboration with researchers at UCLA and UCSB

All The Paws – Founder | East Orange, NJ | Sep 2025 – Present

- Founded and registered a pet services business, growing to 15+ recurring clients within months; identified operational data patterns that directly informed the development of PawsIQ.
- Designed and implemented end-to-end booking and scheduling systems covering invoicing, daily care routines, and emergency protocols, reducing operational friction across all client accounts.
- Built and actively maintain the business website, managing branding, service copy, and client-facing UX to support client acquisition.

EDUCATION

Data Science B.S. | New Jersey Institute of Technology, Newark, NJ | Expected Dec 2027

Coursework: Intro to ML, Regression Analysis, Statistical Methods I & II, Statistical Methods in Data Science, Multivariate Distribution, Linear Algebra
Study Abroad | Hanyang University ERICA, Ansan, South Korea | Feb 2025 – Jun 2025

HONORS & ACTIVITIES

Onyx Dance Crew Choreographer, Social Media Coordinator | NJIT | 2025
Headstarter AI Software Engineering Fellow | 2024
Community Service Certificate | South Korea | 2024
JPMorgan Chase Career.edYOU Academy Graduate | 2023

CERTIFICATIONS

Python Certification (In Progress) | freeCodeCamp | 2026
Supervised Machine Learning: Regression and Classification | Coursera | 2025
Foundations: Data, Data, Everywhere | Coursera | 2024